

HTML Fundamentals

Study Notes — Web Development Basics

1. IDs and Classes

ID Attribute

Used to uniquely identify a single element on the page. No two elements should share the same ID.

```
<p id="txt">Lorem ipsum</p>
<p id="txt_1">Another paragraph</p>
```

- IDs must be unique per page.
- Used with CSS (#id) and JavaScript (getElementById).

Class Attribute

Used to group multiple elements so they can share the same styling or behavior.

```
<p class="txt_1">...</p>
<h1 class="txt_1">...</h1>
```

- Multiple elements can share the same class.
- Used with CSS (.class) and JavaScript (getElementsByClassName).

2. Block vs Inline Elements

Type	Elements	Behaviour
Block	<p>, <h1>, <div>, , 	Takes full width; starts on a new line
Inline	, , 	Takes only as much width as needed; flows within text

3. The <div> Element

A block-level container used to group elements and apply shared styles.

```
<div style="background-color: aqua;">
  <p>Content here</p>
</div>
```

- Inline styles use the style attribute directly on the tag.
- Colors can be specified as names (aqua, brown) or rgb() values.

4. Lists

```
<ul> – Unordered list (bullet points)
<ol> – Ordered list (numbers)
<li> – List item (used inside both)
```

5. Embedding Media

Tag	Purpose	Key Attributes
<iframe>	Embed external pages (Google Maps, YouTube)	src, width, height, allowfullscreen
<audio>	Embed audio files	src, controls, loop, autoplay
<video>	Embed video files	src, controls, loop, width

6. Semantic HTML5 Layout Tags

These tags give meaning to the structure of a webpage (vs generic <div>):

Tag	Purpose
<header>	Top section — logo, site title, navigation
<nav>	Navigation links / menus
<section>	Thematic content grouping
<aside>	Sidebar content — related info, ads
<footer>	Bottom section — copyright, contact info

7. HTML Special Character Entities

Special characters must be escaped in HTML using entity codes. Each entity starts with & and ends with ;

Symbol	Name	Entity Name	Entity Number
<	Less-than sign	<	<
>	Greater-than sign	>	>
&	Ampersand	&	&
"	Quotation mark	"	"
¢	Cent sign	¢	¢
£	Pound sign	£	£
¥	Yen sign	¥	¥
€	Euro sign	€	€
©	Copyright	©	©
®	Registered trademark	®	®
TM	Trademark	™	™
x	Multiplication sign	×	×
//	Division sign	÷	÷
'	Left single quote	‘	‘
'	Right single quote	’	’
"	Left double quotes	“	“
"	Right double quotes	”	”

Tip: Use semantic tags for better SEO and accessibility. Always validate your HTML at validator.w3.org

HTML <input type="url">

Study Notes — URL Input Field

1. What is input type="url"?

The <input type="url"> element creates a text field specifically designed to accept a web address (URL). The browser automatically validates the format before the form is submitted.

Basic Syntax (from your code):

```
<form action="example.com.pk" method="get">

  <label for="URL">
    Please Enter Your Profile URL:
    <input type="url" name="url" id="URL">
  </label>

  <button>Submit</button>
</form>
```

Note: This form was inside HTML comment tags (<!-- -->) in your file, so it was disabled. Remove the comment tags to activate it.

2. Built-in Browser Validation

The browser automatically checks that the entered value looks like a valid URL. It must include a protocol such as http:// or https://. If invalid, the form will NOT submit and the browser shows an error message automatically — no JavaScript needed!

Input Value	Valid?	Reason
https://google.com	YES	Has protocol + domain
http://mysite.pk	YES	Has protocol + domain
google.com	NO	Missing http:// or https://
just some text	NO	Not a URL format at all

3. Useful Attributes

Attribute	Example	Purpose
placeholder	placeholder="https://example.com"	Hint text shown inside the empty field
required	required	Makes the field mandatory before submit
pattern	pattern="https://.*"	Custom regex for extra validation
value	value="https://mysite.com"	Pre-fills the field with a default URL
maxlength	maxlength="200"	Limits the number of characters
size	size="40"	Controls the visible width of the field
disabled	disabled	Makes the field uneditable (greyed out)
readonly	readonly	User can see but not change the value

4. Full Example with Best Practices

```
<form action="example.com.pk" method="get">

  <label for="URL">Please Enter Your Profile URL:</label>

  <input
    type="url"
    name="url"
    id="URL"
    placeholder="https://yourprofile.com"
    required
    maxlength="300"
  >

  <button type="submit">Submit</button>
</form>
```

5. Mobile Keyboard Behaviour

On smartphones and tablets, `type="url"` triggers a special keyboard that includes shortcut keys making it easier to type URLs:

Feature	Detail
.com button	One tap to add .com suffix
/ key visible	Easy to type path separators
No space bar	Spaces are not valid in URLs — keyboard hides or restricts it
https:// shortcuts	Some browsers pre-fill the protocol

6. `type="url"` vs `type="text"`

Both look the same visually, but `type="url"` adds these benefits automatically:

Feature	<code>type="text"</code>	<code>type="url"</code>
Format validation	NO	YES — must be a valid URL
Mobile keyboard	Standard keyboard	URL-optimised keyboard
Browser autofill	Generic	Suggests saved URLs
Semantic meaning	None	Tells browser it expects a URL

Quick Tip: Always pair `<input>` with a `<label>` using matching `for` and `id` values — it improves accessibility and makes the label clickable.